

Exercise 1

1. Which of these sentences are propositions? What are the truth values of those that are propositions?

- a) Miami is the capital of Florida.
- b) $2 + 3 = 5$.
- c) $5 + 7 = 10$.
- d) $x + 2 = 11$.
- e) Answer this question.
- f) The moon is made of green cheese.
- g) $2n \geq 100$.

2. What is the negation of each of these propositions?

- a) $2 + 1 = 3$.
- b) The summer in Maine is hot and sunny.
- c) Jennifer and Teja are friends.
- d) Abby sent more than 100 text messages every day.
- e) 121 is a perfect square.
- f) There is no pollution in New Jersey.
- g) There are 13 items in a baker's dozen.

3. Let p , q , and r be the propositions

p : You have the flu.

q : You miss the final examination.

r : You pass the course.

Express each of these propositions as an English sentence.

- a) $p \rightarrow q$.
- b) $\neg q \leftrightarrow r$.
- c) $q \rightarrow \neg r$.
- d) $p \vee q \vee r$.
- e) $(p \wedge q) \vee (\neg q \wedge r)$.

4. Let p , q , and r be the propositions

p : You get an A on the final exam.

q : You do every exercise in this book.

r : You get an A in this class.

Write these propositions using p , q , and r and logical connectives (including negations).

- a) You get an A in this class, but you do not do every exercise in this book.
- b) You get an A on the final, you do every exercise in this book, and you get an A in this class.
- c) To get an A in this class, it is necessary for you to get an A on the final.
- d) You get an A on the final, but you don't do every exercise in this book; nevertheless, you get an A in this class.
- e) Getting an A on the final and doing every exercise in this book is sufficient for getting an A in this class.
- f) You will get an A in this class if and only if you either do every exercise in this book or you get an A on the final.

5. Determine whether these biconditionals are true or false:

- a) $2 + 2 = 4$ if and only if $1 + 1 = 2$.
- b) $1 + 1 = 2$ if and only if $2 + 3 = 4$.
- c) $1 + 1 = 3$ if and only if monkeys can fly.
- d) $0 > 1$ if and only if $2 > 1$.

6. Determine whether each of these conditional statements is true or false:

- a) If $1 + 1 = 2$, then $2 + 2 = 5$.
- b) If $1 + 1 = 3$, then $2 + 2 = 4$.
- c) If $1 + 1 = 3$, then $2 + 2 = 5$.
- d) If monkeys can fly, then $1 + 1 = 3$.

7. Construct a truth table for each of these compound propositions:

- a) $p \wedge \neg p$.
- b) $p \vee \neg p$.
- c) $p \rightarrow \neg p$.
- d) $p \leftrightarrow \neg p$.
- e) $(p \vee \neg q) \rightarrow q$.
- f) $(p \vee q) \rightarrow (p \wedge q)$.
- g) $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$.
- h) $(p \rightarrow q) \rightarrow (q \rightarrow p)$.
- i) $(p \wedge q) \rightarrow (p \vee q)$.
- j) $p \oplus (p \vee q)$.
- k) $(p \vee q) \rightarrow (p \oplus q)$.
- l) $(p \oplus q) \rightarrow (p \oplus \neg q)$.
- m) $p \rightarrow (\neg q \vee r)$.
- n) $(p \rightarrow q) \wedge (\neg p \rightarrow r)$.
- o) $\neg p \rightarrow (q \rightarrow r)$.
- p) $(p \oplus q) \rightarrow \neg r$.

8. What is the value of x after each of these statements is encountered in a computer program, if $x = 1$ before the statement is reached?

- a) **if** $x + 2 = 3$ **then** $x := x + 1$.
- b) **if** $(x + 1 = 3)$ **OR** $(2x + 2 = 3)$ **then** $x := x + 1$.
- c) **if** $(2x + 3 = 5)$ **AND** $(3x + 4 = 7)$ **then** $x := x + 1$.
- d) **if** $(x + 1 = 2)$ **XOR** $(x + 2 = 3)$ **then** $x := x + 1$.
- e) **if** $x < 2$ **then** $x := x + 1$.

9. Use truth tables to verify these equivalences:

- a) $p \wedge T \equiv p$
- b) $p \wedge F \equiv F$
- c) $p \wedge q \equiv q \wedge p$.
- d) $\neg(p \wedge q) \equiv \neg p \vee \neg q$.
- e) $(p \vee q) \vee r \equiv p \vee (q \vee r)$.
- f) $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.

10. Show that each of these conditional statements is a tautology by using truth tables:

- a) $(p \wedge q) \rightarrow p$
- b) $\neg p \rightarrow (p \rightarrow q)$
- c) $(p \wedge q) \rightarrow (p \rightarrow q)$
- d) $\neg(p \rightarrow q) \rightarrow \neg q$

11. Show that each conditional statement in Exercise 10 is a tautology without using truth tables.

12. Let $P(x)$ denote the statement " $x \leq 4$." What are these truth values?

- a) $P(0)$
- b) $P(4)$
- c) $P(6)$

13. Determine the truth value of each of these statements if the domain consists of all integers:

- a) $\forall n(n + 1 > n)$
- b) $\exists n(2n = 3n)$
- c) $\exists n(n = -n)$
- d) $\forall n(3n \leq 4n)$

14. Determine the truth value of each of these statements if the domain for all variables consists of all integers:

- a) $\forall n(n^2 \geq 0)$
- b) $\exists n(n^2 = 2)$
- c) $\forall n(n^2 \geq n)$
- d) $\exists n(n^2 < 0)$

15. Use a direct proof to show that the sum of two odd integers is even.

16. Show that the additive inverse, or negative, of an even number is an even number using a direct proof.

17. Prove that if $m + n$ and $n + p$ are even integers, where m , n , and p are integers, then $m + p$ is even. What kind of proof did you use?

18. Use a direct proof to show that every odd integer is the difference of two squares.

19. Show that if n is an integer and $n^3 + 5$ is odd, then n is even using:

- a) a proof by contraposition.
- b) a proof by contradiction.

20. Prove that if m and n are integers and mn is even, then m is even or n is even.